

Field and Galois Theory

(Final Test, BMath-3rd year, 2025)

Instructions: Total time 3 Hour. Solve any **FIVE** problems for a max score of 60. Use terminologies, notations and results as covered in the course, no need to prove such results. If you wish to use a problem given in a homework or an exercise from the class, please supply its full solution.

1. Let E/F be a cyclic field extension with $[E : F] = n$. Let $d|n$ and $n = dm$ with $m > 1$. Suppose $c \in F^\times$ is such that $c^d \in N_{E/F}(E)$. Show that there exists y in the unique subfield K of E/F with $[K : F] = d$ such that, $c = N_{K/F}(y)$. (12)
2. Let k be a field of characteristic 0. Let L/k be a separable field extension of degree 3. Assume L/k is not Galois. Prove that there is a quadratic field extension D of k such that the compositum LD is a cyclic extension of D . (12)
3. Let $f(X) \in \mathbb{Z}[X]$ be a degree 4 irreducible polynomial with Galois group S_4 over $\mathbb{Q}$. Let $\alpha \in \mathbb{C}$ be a root of f . Prove that $\mathbb{Q}(\alpha)/\mathbb{Q}$ has no subextensions of degree 2. (12)
4. Let F be a finite field and $f(X)$ be an irreducible polynomial of degree 3 in $F[X]$. Show that the discriminant of $f(X)$ is a square in F . (12)
5. Let K be a finite field extension of degree n of a finite field F , with $|F| = q$ and let $d \in F^\times$ be fixed. **(i)** Prove $N_{K/F} : K \rightarrow F$ is surjective. **(ii)** Let $K_d = \{a \in K \mid N_{K/F}(a) = d\}$. Determine $|K_d|$. (6+6)
6. For $a, b, c, d, e \in \mathbb{Q}$, show that the polynomial $aX^8 + bX^6 + cX^4 + dX^2 + e \in \mathbb{Q}[X]$ is solvable by radicals. (12)